

BAYAN, Mongolia

WMO: 442940

Lat: 47.2567N

Lon: 107.5378E

Elev: 4692

StdP: 12.37

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-29.3	-25.4	-35.2	0.9	-28.4	-31.4	1.1	-24.5	26.8	-4.5	23.2	-4.4	3.3	0	0.735

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.8	85.3	60.7	81.4	59.3	77.9	57.9	63.9	78.9	62.0	76.1	60.4	73.4	9.5	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.1	89.1	68.4	57.1	82.8	67.5	55.2	77.2	66.2	31.8	78.4	30.3	76.2	29.0	73.6	72.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	27.6	23.3	21.4	DB	-35.0	90.9	4.8	4.3	-38.4	94.0	-41.2	96.5	-43.9	98.9	-47.4	102.1
(5)				WB	-34.8	67.6	4.7	3.4	-38.2	70.0	-41.0	72.0	-43.6	73.9	-47.1	76.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	30.3	-7.9	-0.6	16.8	34.9	48.3	60.1	65.1	60.6	48.3	31.2	9.1	-4.1	(6)
(7)		DBStd	27.20	7.87	9.74	11.28	9.33	8.78	7.29	6.09	6.74	8.39	8.90	12.20	8.73	(7)
(8)		HDD50	8469	1796	1418	1029	458	138	9	0	5	129	583	1227	1677	(8)
(9)		HDD65	12805	2261	1838	1494	902	520	180	72	169	502	1048	1677	2142	(9)
(10)		CDD50	1283	0	0	0	5	86	311	469	334	78	0	0	0	(10)
(11)		CDD65	145	0	0	0	0	3	33	76	32	1	0	0	0	(11)
(12)		CDH74	1800	0	0	0	1	99	478	804	387	31	0	0	0	(12)
(13)		CDH80	487	0	0	0	0	17	131	242	95	2	0	0	0	(13)
(14)	Wind	WSAvg	8.0	6.0	6.6	8.6	10.7	11.7	9.6	8.3	7.5	8.1	7.6	6.2	5.7	(14)
(15)	Precipitation	PrecAvg	8.8	0.0	0.0	0.1	0.2	0.6	1.7	2.6	2.3	0.9	0.2	0.2	0.0	(15)
(16)		PrecMax	14.5	0.2	0.3	0.4	1.0	3.4	7.2	5.2	7.8	2.9	0.7	0.6	0.2	(16)
(17)		PrecMin	5.2	0.0	0.0	0.0	0.0	0.0	0.0	0.7	0.3	0.0	0.0	0.0	0.0	(17)
(18)		PrecStd	2.3	0.0	0.0	0.1	0.2	0.6	1.3	1.2	1.4	0.7	0.2	0.2	0.0	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.0	32.6	52.9	69.7	82.4	88.9	91.5	87.7	77.8	63.0	43.9	22.3	(19)
MCWB			17.4	25.8	37.6	48.0	53.9	60.3	64.6	62.9	55.4	44.7	34.3	18.9	(20)	
2%		DB	14.5	27.1	46.0	62.6	76.6	83.8	86.3	82.0	73.2	56.9	38.1	17.2	(21)	
		MCWB	11.9	21.5	34.1	43.9	53.4	58.1	62.7	60.1	54.2	43.0	30.6	14.6	(22)	
5%		DB	10.0	22.1	41.0	57.7	72.0	79.3	81.8	78.0	69.0	52.4	33.1	13.5	(23)	
		MCWB	8.0	18.2	31.2	41.9	50.5	56.5	61.0	58.9	52.1	39.9	27.4	11.5	(24)	
10%		DB	5.8	17.2	35.8	53.1	67.4	75.4	78.2	74.4	65.0	48.0	28.4	10.0	(25)	
		MCWB	4.3	14.4	28.0	39.4	48.5	55.6	59.7	57.5	50.2	37.4	24.0	8.3	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.5	26.2	38.3	50.4	59.1	63.8	68.3	65.3	58.8	47.0	34.7	19.3	(27)
MDCB			20.9	32.0	50.9	66.9	77.1	80.1	85.3	79.8	71.3	60.4	43.8	22.2	(28)	
2%		WB	12.1	21.5	34.4	45.2	54.8	61.1	65.3	62.7	55.6	43.3	30.8	14.9	(29)	
		MDCB	14.6	26.5	45.1	59.8	72.4	77.9	80.8	77.5	69.8	55.1	37.4	17.1	(30)	
5%		WB	8.1	18.3	31.3	42.3	51.9	59.2	63.0	60.9	53.3	40.5	27.6	11.7	(31)	
		MDCB	9.9	22.0	40.4	56.4	68.5	74.7	77.1	73.6	66.4	51.8	32.9	13.6	(32)	
10%		WB	4.3	14.4	28.1	39.6	49.2	57.2	61.4	59.1	51.0	37.7	24.0	8.4	(33)	
		MDCB	5.8	17.3	35.5	52.8	64.9	70.9	74.3	71.4	63.5	47.3	28.4	10.0	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	23.2	26.8	26.4	26.0	26.0	22.8	20.8	22.1	25.6	25.4	23.8	21.6	(35)
MCDBR			28.3	32.3	32.2	33.3	34.5	29.0	26.7	28.1	31.9	33.2	28.9	23.9	(36)	
MCWBR			25.8	27.5	22.1	20.0	18.2	12.8	11.0	12.3	16.3	21.2	22.3	21.5	(37)	
5% WB		MCDBR	28.0	31.7	31.8	31.6	30.8	25.4	23.0	24.0	27.8	31.6	28.3	23.7	(38)	
		MCWBR	25.5	27.2	22.3	20.3	18.4	13.0	11.3	12.1	15.3	20.6	22.1	21.4	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.236	0.260	0.298	0.345	0.379	0.392	0.372	0.352	0.310	0.280	0.251	0.237	(40)	
(41)		taud	2.368	2.325	2.286	2.223	2.166	2.228	2.344	2.376	2.432	2.457	2.413	2.362	(41)	
(42)		Ebn at Noon	278	292	293	287	280	276	281	282	287	281	269	263	(42)	
(43)		Edh at Noon	22	28	35	41	45	43	38	35	30	25	21	19	(43)	
(44)	All-Sky Solar Radiation	RadAvg	629	991	1466	1864	2160	2132	2010	1804	1519	1062	671	500	(44)	
(45)		RadStd	44	66	55	77	130	115	107	104	73	54	40	35	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	-2.65	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (4 neighbors)	N/A	N/A	-3.83	N/A	-1.22	-1.17	N/A	N/A	N/A	N/A	

Nomenclature: See separate page